

On the Substitution of Distribution Entanglers: The Commutator Criterion and Directional Collapse Extension 11

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Abstract

The baseline Q-LINK architecture utilizes messenger-controlled CNOT gates for its distribution layer (U_{dist}). This note formally establishes the algebraic constraints for substituting this entangler. By evaluating the necessary conditions for Dynamical Lie Algebra (DLA) expansion, we establish the ‘‘Commutator Criterion,’’ defining the exact viable operator basis for replacement entanglers (e.g., CZ , CY). Furthermore, we prove the ‘‘Directionality Theorem’’: if the distribution entangler is orientation-reversed (e.g., data-controlled CNOTs targeting the messenger), the DLA suffers a fatal fracture, effectively severing the coherent residual pathway.

1 The Commutator Criterion for Substitution

Let the collection layer utilize an interaction basis defined by the generator $G_{coll} = Q_m \otimes A_i$, where $Q_m \in \{X, Y, Z\}$ is a Pauli acting on the messenger qubit. For the standard R_{xx} architecture, $Q_m = X_m$.

We propose replacing the distribution CNOT gates with an arbitrary entangler whose dominant Lie generator is $G_{dist} = P_m \otimes B_i$. The expansion of the DLA depends entirely on the Lie bracket between these layers:

$$[Q_m \otimes A_i, P_m \otimes B_i] = (Q_m P_m) \otimes (A_i B_i) - (P_m Q_m) \otimes (B_i A_i) \quad (1)$$

If P_m commutes with Q_m (i.e., $P_m = Q_m$), the bracket evaluates to:

$$Q_m^2 \otimes [A_i, B_i] = I_m \otimes [A_i, B_i] \quad (2)$$

As established in prior ablations, generating I_m strictly closes the algebra within a fractured subspace, trapping the messenger and destroying the architectural advantage.

First Axiom of Substitution: To preserve the full $\mathfrak{su}(2^{n+1})$ expansion, the replacement distribution entangler must project a Pauli basis onto the messenger (P_m) that strictly anti-commutes with the collection basis (Q_m).

1.1 Viable Isomorphic Substitutions

Assuming an R_{xx} collection layer ($Q_m = X_m$), the distribution entangler must project Z_m or Y_m .

- **Control-Z (CZ):** Generator $\propto Z_m \otimes Z_i$.
 $[X_m \otimes X_i, Z_m \otimes Z_i] = -2(Y_m \otimes Y_i)$. The algebra successfully expands.
- **Control-Y (CY):** Generator $\propto Z_m \otimes Y_i$.
 $[X_m \otimes X_i, Z_m \otimes Y_i] = 2(Y_m \otimes Z_i)$. The algebra successfully expands.

Both CZ and CY gates (controlled by the messenger) are mathematically perfect isomorphic substitutes for the CNOT.

2 The Directionality Theorem (Target vs. Control)

A critical vulnerability emerges when evaluating the physical orientation of the entangler. Suppose we reverse the direction of the standard CNOT, such that the data qubit i is the control, and the messenger m is the target.

The Lie generator of this reversed CNOT is proportional to $Z_i \otimes X_m = X_m \otimes Z_i$. We evaluate its commutator against the R_{xx} collection generator ($X_m \otimes X_i$):

$$[X_m \otimes X_i, X_m \otimes Z_i] = X_m^2 \otimes [X_i, Z_i] = I_m \otimes (-2iY_i) \quad (3)$$

Because the X -target of the reversed CNOT perfectly commutes with the X -basis of the collection layer, the commutator yields the identity operator I_m on the messenger subspace.

Theorem of Directional Collapse: The DLA of an orientation-reversed distribution layer strictly closes without ever generating Y_m or Z_m . The messenger qubit becomes algebraically trapped in the X -eigenbasis. Consequently, it acts as a static, incoherent phase sink rather than a dynamic residual conduit, and the geometric barren plateau advantage mathematically vanishes.

3 Conclusion

The replacement of the distribution entanglers in the Q-LINK architecture is highly constrained. While alternative topologies like CZ or CY are perfectly viable, the substitution must rigorously obey the Commutator Criterion. More critically, the “direction” of the residual pathway is not merely a conceptual metaphor, but an absolute algebraic law: the messenger qubit must operate as the strictly non-commuting *control* operator in the distribution phase to prevent catastrophic DLA fracturing.